

Are Serious Games Engaging? A Case of Motivating Sustainable Household Practices

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Abstract

Serious games aim to capitalize on game qualities, such as engagement and entertainment, to achieve a serious purpose. Through interviews from two evaluations of a serious game about environmentally sustainable household practices, this study investigates whether the 28 participants experienced the game as engaging and entertaining. We find (a) that the participants merely tended toward being positively engaged by the game, (b) that it was experienced as more engaging than entertaining, and (c) that the participants were not positively entertained by it. In addition, several participants appeared more engaged in the underlying topic of sustainability than in the gameplay, that is, they favored relatability and relevance over entertainment. On that basis, we conclude that serious games may not need to be entertaining to fulfil their serious purpose.

CCS Concepts

• **Human-centered Computing**; • **Human-computer Interaction (HCI)**; • **Empirical studies in HCI**;

Keywords

Serious games, Engagement, Entertainment, Sustainability, Household practices

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1 Introduction

Serious games have “an explicit and carefully thought-out educational purpose and are not intended to be played primarily for amusement” [1, 9]. Instead, the game qualities of engagement and entertainment are employed to serve the serious purpose. Serious games are, for example, used to inform people about the environmental consequences of everyday practices and nudge them toward adopting more sustainable practices [2, 18, 28, 34, 38]. Evidently, this purpose is external to the game itself. To achieve it, a serious game must have enough impact to change the player’s thinking

and doing in ways that reach beyond the gameplay. In this study, we investigate the engagement and entertainment experienced by the players of a serious game intended to motivate the adoption of environmentally sustainable household practices.

Engagement and entertainment are central game qualities, often mentioned as defining characteristics of games, including serious games [4, 20, 39]. *Engagement* boosts motivation, which is important to the purpose of a serious game [20]. Though serious games are different from entertainment games, that is, games played primarily for amusement purposes [1], *entertainment* has been linked with learning through terms such as edutainment [23]. The game investigated in this study was designed with the long-term goal of impacting the player’s everyday practices, through a carry-over effect from playing the game. The game qualities of engagement and entertainment are intended to support this goal. They form the rationale for designing a serious game, rather than, say, a pamphlet that only provides information. On this basis, we ask the research question: *Are serious games experienced as engaging and entertaining?* Our main contribution is that the studied game is significantly more engaging than entertaining. It may be misconstrued to expect serious games to be entertaining by default.

2 Related Work

Since Abt coined the term ‘Serious Games’ in his 1970-book [1], serious games have evolved into a global, billion-euro industry [9, 11]. It encompasses both analog and digital games [10], targets a variety of audiences [6, 15, 19, 28], and serves a wide spectrum of purposes, such as supporting formal education [12, 40], organizational upskilling [7, 21], personal life-adjustments [5, 15, 16] and behavior change [6, 34]. Abt’s definition of serious games is from an analog era, yet continues to form the basis of newer definitions, including Michael & Chen [25] and de Freitas & Liarokapis [11].

The dictionary understanding of the word ‘game’ often hinges on the game’s purpose of fun and entertainment [41, 42]. However, McGonigal [24] provides a more design-oriented understanding of games, emphasizing four functional characteristics: an in-game goal, rules, a feedback system, and voluntary participation. Though McGonigal [24] does not explicitly mention fun, amusement, or entertainment, her definition emphasizes how structured elements promote a sense of purpose, unleash creativity, and provide motivation—factors often associated with engagement and entertainment. Michael & Chen [25] adopt a more experience-based



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view, with additional characteristics like attention and creating social groups, both resonating well with the concepts of engagement [26.] and entertainment [32.].

The question of whether serious games *need* to be fun remains debated [25.]. Many game definitions emphasize elements that can foster enjoyable and engaging experiences. Reflecting this, the game industry has long applied design elements to enhance player motivation, enjoyment, and engagement. This brings us to a critical distinction in serious-game experiences: the difference between engagement and entertainment. Drawing on Mills et al. [26.], we understand *engagement* as a meta-construct comprising affective, behavioral, and cognitive components—including effort, persistence, emotional arousal, interest, attention, and concentration. In contrast, *entertainment*, as defined by Savolainen [32.], is characterized by psychological relaxation, stimulation, fun, joy, and a merry atmosphere. This distinction is especially relevant in the context of serious games, where the intended learning or behavioral outcome often requires active cognitive and emotional investment rather than merely passive enjoyment [26.]. While entertainment can support initial interest or ease of access [30.], sustained engagement is more closely tied to the achievement of serious purposes such as skill acquisition or behavior change [6., 30.]. As such, serious games that succeed in engaging players may serve their purpose without being entertaining in the traditional sense. Michael & Chen [25.] positions entertainment as a possible game outcome rather than a necessary game component. Still, entertainment cannot be discounted, as recognized by Abt, who, in direct continuation of his previously cited definition, clarifies: “This does not mean that serious games are not, or should not be, entertaining.” [1.:9.]. In short, engagement and entertainment may not be the primary goals, but they remain central to the understanding of serious games.

This dual understanding of serious games as a vehicle for both entertainment and serious purposes is often invoked as a reason for their value as tools for learning and behavior change [14., 36.]. In the field of environmental sustainability [29., 31., 36.], this argument is frequently combined with the assumption that the simulated game setting provides a sense of psychological safety, enabling players to explore, experiment, and internalize new knowledge and skills [14., 28., 34.]. Consequently, serious games have been widely employed as behavior-change interventions in both private [37.] and organizational contexts [33.], addressing a variety of environmental sustainability challenges, from resource conservation [3., 31.] over transportation [8., 13.], to waste management [17., 35.].

3 Method

This study is based on interviews from two evaluations of a functional serious-game prototype intended for households, themed around sustainable water consumption. The game is text-based and presents the player with dilemma scenarios, within a narrative setting where the player is cast as the head of a four-person household including a partner, a teenaged daughter and a pre-teen son. Each dilemma contains three or four choices, of which the player must choose one. The choice elicits choice-specific feedback and a visualized point score, before the player proceeds to the next dilemma.

The two evaluations occurred during the game development process. The first evaluation (October 2023) followed a discount usability-engineering approach [27.]. Five participants were invited to play the game in individual sessions, while simultaneously being interviewed about their experience. With the participants’ consent, the sessions were documented through video, audio, and screen recordings. The participants were sourced from the personal and professional networks of the project group members. The second evaluation (February 2024) was conducted during a municipal climate-awareness event. Eventgoers were invited to participate by first playing the game on one of five provided laptops and then being interviewed about their experience. While the play-through was unfacilitated, the interview was conducted by one of three interviewers. This evaluation had 23 participants. The online appendix provides an overview of the 28 participants in the two evaluations.

All 28 participants gave their oral consent to participate and be recorded. The audio-recordings of the interviews were transcribed and analyzed by the authors in collaboration, using a six-step content-analysis process [22.]. First, the transcripts were segmented into units of meaning, each a few sentences long. The segmentation divided the transcripts into 1919 such units. Second, four randomly selected interviews were coded by all three authors independently. Each unit of meaning was coded for whether the participants expressed being *engaged* in playing the game and whether they experienced the game as *entertaining*. We used the definitions of engagement and entertainment proposed by Mills et al. [26.] and Savolainen [32.], respectively. For both codes, each instance was categorized as ‘Positive’, ‘Negative’, or ‘Neutral’. Third, the authors met to discuss all disagreements in their coding until consensus. This step served to create a shared understanding of the coding categories. Fourth, six of the remaining interviews were randomly selected and coded by two of the authors independently, each author coding four interviews. Fifth, the authors met to discuss and reach consensus about all disagreements in their coding. The kappa values of the agreement among the authors in their coding of these six interviews were 0.67 (engaging) and 0.74 (entertaining). Both kappa values were above the threshold of 0.60 recommended by Lazar et al. [22.] as indicating satisfactory reliability. Sixth, each author coded six of the remaining 18 interviews.

4 Results

The 28 participants expressed a total of 1919 units of meaning during the interviews, corresponding to a mean of 68.5 units per participant. Table 1 shows the distribution of the meaning units across the engagement and entertainment categories. For each category, the table gives the total number of units in the category across all 28 participants and the mean percentage of units in the category for a single participant. To assign equal weight to participants irrespective of the number of units they expressed, the statistical analyses were performed on these percentages, not on the frequency counts.

Table 1: Count and percentage of meaning units for the 28 participants

	Engagement		Entertainment	
	Count	Percentage ^a	Count	Percentage ^a
Positive	598	24 (± 15)	64	2 (± 3)
Negative	344	17 (± 11)	102	6 (± 6)
Neutral	977	59 (± 15)	1753	92 (± 6)

^a The percentage of meaning units in a category (e.g., positive) was first calculated for each participant and then averaged across participants. The table gives the mean percentage ($\pm$ standard deviation).

4.1 The game merely tended toward being engaging

The participants expressed positive engagement in the game in as much as 24% of the units of meaning and negative engagement (i.e., disengagement) in 17%. In the remaining units of meaning, the participants neutrally described aspects of the game or talked more broadly about their views on sustainability. The percentage of positive-engagement meaning units was not significantly higher than that of negative-engagement units, $t(27) = 1.53$, $p = 0.07$. That is, the participants merely tended toward finding the game engaging. Positive engagement was, for example, expressed like this: “It is just cool with such [informational] details” (Participant 2), “This [piece of information] is good to know, actually” (Participant 3), and “It was informative and inspiring” (Participant 12). More specifically, participants expressed engagement in the sense that they recognized the game dilemmas from their own households and, therefore, experienced them as relatable and relevant. In the words of Participant 18: “I find that some of them [i.e., the dilemmas] are definitely relatable. They relate to things you can consider implementing at home.” Participant 24 was particularly drawn in by the way in which the choices in the game were presented as dilemmas: “It made me think about how I can save water – and about what saves *water* and what saves *money*. [...] Sometimes, it was actually a little difficult to know what the better choice was.”

In contrast, it disengaged Participant 20 that the household in the game reacted very differently from how her own household would react, thereby making it difficult for her to relate to the game: “I chose something and then the game responded that the daughter [in the game household] considered that tiresome. My daughter would not consider that tiresome; she has been brought up differently. That put me off a bit.” Participant 24 also expressed disengagement, but for a different reason. She did not have the patience to read the explanatory text in the game and, thus, skipped through it quite quickly: “I am the type of person who does things quickly. So, I didn’t [read the explanatory text].”

4.2 Engagement exceeded entertainment

While the participants expressed positive engagement in the game in 24% of the meaning units, they expressed positive entertainment in as little as 2%. The percentage of positive-engagement meaning units was significantly higher than that of positive-entertainment units, $t(27) = 8.20$, $p < 0.001$. In fact, none of the 28 participants were entertained more often than engaged. When the participants were entertained, the entertainment often heightened their engagement in the game or its possible uses. For example, Participant 1

explained that he enjoyed the game because it provided entertaining possibilities for making sustainability a conversation topic in his real-life household: “It is quite fun. You can use it to talk with your family [about sustainability]. If you go home and ask them ‘Why are we doing things this way?’, then maybe... I actually find that quite fun.” However, the low percentage of entertaining meaning units indicated that engagement was primarily obtained in other ways than through entertainment. We also found a stronger correlation between the percentage of positive engagement and the time the participants spent playing the game, $r(26) = 0.67$, than between the percentage of positive entertainment and play time, $r(26) = 0.49$. This provided further evidence of engagement being more closely tied than entertainment, to the participants’ interaction with the game.

4.3 The game was not entertaining

As many as 15 of the 28 participants never expressed positive entertainment. Overall, the percentage of negative-entertainment meaning units was significantly higher than that of positive-entertainment units, $t(27) = -2.51$, $p < 0.01$. That is, the participants did not find the game entertaining. For example, Participant 8 largely disregarded the information about her play score and, thereby, did not attend much to the game component that most directly aimed to make the game entertaining: “I didn’t use them [i.e., the visualizations of the play score] very much. No. They were not what motivated me.” Participant 23 reacted against the entire idea of a dilemma-based game. While the dilemmas were intended to foster reflection, increase realism, and make the game entertaining, she would have preferred information about what to do. One of the dilemmas was about different options for washing the family car; the participant said: “It [i.e., the game] does not make sense because I have not been instructed about whether the right thing for me to do is to drive to the eco-certified carwash or to wash my car at home with my water hose. [...] I don’t know that after having played the game.”

4.4 Emergent findings from the coding process

In addition to the predefined codes used in our content analysis, two qualitative patterns emerged during the analysis. The first pattern was a distinction between engagement with the game itself and engagement with the theme of the game. While the former concerned the gameplay and the achieved point score, the latter concerned the underlying topic of sustainability and water consumption. Several participants appeared more engaged in the underlying topic than in the gameplay. For instance, Participant 5 compared her in-game

choices to her real-life household practices: “I have actually done those water-saving actions that are obvious to do. So, it is maybe really more the behavior of the kids and similar, which can be hard to do something about, where you can get the biggest advantage.” These statements were not strictly tied to gameplay engagement but, nevertheless, indicated a meaningful form of thematic involvement. The second pattern related to participants’ perception of the game as a ‘game.’ Despite the integration of several game elements (e.g., the point system, simulated family interactions, and choice-based feedback), some participants questioned whether their experience qualified as gameplay. For example, Participant 2 stated that he was unsure of the game’s overall goal, which undermined one of the fundamental characteristics of a game: “Yes, well, you could say that, as a participant in the game, that is what I said before, you don’t know your [in-game] goal.”

5 Concluding Discussion

This paper investigates whether a serious game is experienced as engaging and entertaining. Our findings show that the participants tended toward experiencing positive engagement, expressed through comments on the relevance and value of the dilemmas. However, they were not entertained, despite game-like features such as a point system and a narrative structure. Hence, the game components that would usually be expected to foster entertainment generally failed to do so. In contrast, engagement was driven by components related to the recognizability and relatability of the narrative and information employed in the game. Several participants expressed interest in the potential of using the game as a group activity and conversation starter with their real-life families.

Our findings raise questions about how the label, *game*, shapes expectations. By definition, a serious game has an educational purpose, rather than being primarily intended for entertainment. However, even when prefixed with ‘serious’, it often evokes associations with fun or enjoyment. In our study, participants did not experience the serious game as entertaining, though several did find it engaging. This dissonance highlights a tension between the formal definition of a serious game and the expectations evoked by the term ‘game’ in everyday language. Our serious game meets the formal definition, with clear rules, feedback mechanisms, narrative structure and voluntary play [24., 25.]. In addition, several participants described being drawn into the gameplay, which resonated with their lived experience, thereby indicating a level of immersion that aligns with the technical definition of a game. However, like all words, ‘game’ is dynamically defined by its current use, which reveals an inherent social expectation that games should be entertaining [41., 42.]. Indeed, Ravyse et al. [30.] find that this quality is critical to the success of a serious games, stating that: “We take the stance that serious games are successful if they get played (granted that the pedagogic infusion is decent)—fun games are played.” They conclude that the players of serious games “want to have fun before they value the subsequent learning-benefit serious games can offer them. Serious games producers must not impede this hunger for fun but rather use it to stealthily engage the player with the required learning material” [30.].

Our findings indicate that the game tended to foster engagement with the topic of sustainability as well as with the game itself. This

way, the game retained the potential to fulfill its serious purpose despite the absent entertainment. The participants’ statements that the game presented sustainability and sustainability dilemmas in relevant and relatable ways are important because they suggest a link from the game to the participants’ household practices. Without such links, it is unlikely that the game will succeed in its serious purpose of changing household practices [5.]. Relevance and relatability would also be required in information pamphlets and other sustainability interventions. As such, they are not specific to serious games in the way the expectation of entertainment is. When examining the stance taken by Ravyse et al. [30.], it also becomes clear that while they consider entertainment critical to the success of the serious game, they tie this importance to whether the game will be played and not directly to its serious purpose. Thus, entertainment may add value to the game but not be instrumental in achieving changes that reach beyond the game. Our data support this, with engagement displayed through multiple instances of participants relating the content of the game with their real life, thereby suggesting that the game could be successful in initiating and supporting behavior change to more sustainable household practices. Thematic relevance, recognizability, and personal resonance contributed to fostering engagement, possibly more than the entertainment-directed game elements did. In appraising the value of engagement and entertainment, the designers and researchers of serious games should carefully consider to what extent engagement can favorably be achieved through means other than entertainment.

There are three limitations to this study. First, our results are specific to the studied game. Future studies should validate our findings for serious games with other game elements and with a serious purpose other than sustainability. Second, the study participants, especially those in the second evaluation, may have been favorably predisposed toward a game about sustainable household practices. At least, their preexisting interest in climate issues was sufficiently strong to make them attend the climate-awareness event that hosted the second evaluation. Future studies should target participants without such a pre-study interest in the serious purpose of the game. Third, it remains an assumption that a serious game will be more effective at stimulating behavior change if it is experienced as engaging. We have investigated the participants’ experience of the game. It is for future work to determine the effect of the game experience on day-to-day household practices.

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